Kodai Kawamura

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Research Interests

My research focuses on probabilistic machine learning and deep generative models, particularly understanding and improving diffusion models through probabilistic modeling. Current applications include language generation and scientific data reconstruction.

Education

- **National University of Singapore** *August 2025 – 2030 (Expected)*
  Ph.D. in Computer Science Singapore
- **Tokyo University of Science** *April 2020 – March 2025*
  B.E. in Computer Science Tokyo, Japan
  - Outstanding Student Award; Dean's Award.
- **University of California, Davis** *April 2022 – March 2023*
  Exchange Program CA, United States

Research Experience

- **National University of Singapore** *August 2025 – Present*
  Ph.D. Research, Advisors: Prof. Kenji Kawaguchi and Prof. Anji Liu Singapore
  - Developed a training-free soft-token decoding method for diffusion language models and analyzed its behavior from a probabilistic modeling perspective. Manuscript under review at ICLR 2027.
  - Investigating diffusion models for reconstructing metal–organic framework (MOF) data.
- **National Institute of Advanced Industrial Science and Technology (AIST)** *June 2024 – August 2025*
  Research Member, Mentor: Dr. Hirokatsu Kataoka Ibaraki, Japan
  - Led a project introducing Approximate Domain Unlearning, a task for selectively removing domain-specific knowledge from pretrained vision-language models.
  - Accepted at NeurIPS 2025 as a Spotlight paper.
- **Korea University** *August 2024 – November 2024*
  Research Intern, Advisor: Prof. Sangpil Kim Seoul, South Korea
  - Worked on watermarking video diffusion models in collaboration with Google DeepMind.
- **Tokyo University of Science** *June 2023 – May 2024*
  Bachelor's Thesis Research, Advisor: Prof. Go Irie Tokyo, Japan
  - Led a project on one-shot test-time adaptation to improve pretrained vision-language classification model using a single unlabeled test sample.
  - Accepted at BMVC 2024.

Publications

[1] **Kodai Kawamura**, Kenji Kawaguchi, Anji Liu. **Rethinking Soft Tokens for Parallel Decoding in Diffusion Language Models**. *ICLR 2027 (Under review)*.

[2] **Kodai Kawamura***, Yuta Goto*, Rintaro Yanagi, Hirokatsu Kataoka, Go Irie. **Approximate Domain Unlearning for Vision-Language Models**. *NeurIPS 2025 (Spotlight)*.

[3] **Kodai Kawamura**, Shunya Yamagami, Go Irie. **Region-based Entropy Separation for One-shot Test-Time Adaptation**. *BMVC 2024*.

* Equal contribution.

Invited Talks and Presentations

- **Forgetting Visual Domains in Vision-Language Models** *October 2025*
  Invited poster presentation at LIMIT Workshop (ICCV 2025)
- **Region-based Entropy Separation for One-shot Test-Time Adaptation** *November 2024*
  Visual Geometry Group (VGG), University of Oxford